



# Computer Graphics 2 - Exercise Sheet 4

Technische Universität Berlin - Computer Graphics

**Handout** 25. June 2026, **Practice Due** 15. July 2026, **Theory Due** 15. July 2026 11:59 pm

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## Exercise 1: Mesh smoothing (3 + 5.5 Points)

In this exercise you will continue working on triangle mesh and implement techniques for mesh smoothing. Detailed tasks are:

1. Implement mesh smoothing using the graph-Laplacian (also called uniform Laplacian) operator: Each vertex is moved with respect to its 1-ring neighbors weighted equally. (1 Point)
2. Implement the cotan-Laplace operator. And calculate the bayrcentric lumped mass matrix. (1 Point)  
You can also calculate the circumcentric dual mass matrix. (1 Bonus Point)
3. Implement mesh smoothing using the cotan-Laplacian following the explicit Euler method. Demonstrate in the presentation, that the procedure is unstable with large step size. (1 Point)
4. Implement mesh smoothing using cotan-Laplacian following the implicit Euler method. (1 Bonus point)  
Implement mesh smoothing using cotan-Laplacian following the implicit trapezoidal rule. (0.5 Bonus point)
5. Computer eigenvectors of cotan-Laplace operators and approximate triangle meshes using different number of (i.e. selectable) eigenvectors. (1 Bonus point)

Smoothing should be visually noticeable. It is completely OK to use meshes generated from the marching cubes method in the last exercise, i.e. you can but don't have to extend your available framework in order to read meshes in other formats e.g. with the help of an external library.

Bonus exercise: Implement Loop<sup>1</sup> or  $\sqrt{3}$  subdivision<sup>2</sup> scheme to refine meshes. (each 1 bonus point)

## Exercise 2: Theory (2 + 4 Points)

1. Please show

$$\frac{\|e_{ij}^*\|_{\pm}}{\|e_{ij}\|} = \frac{1}{2} (\cot(\alpha_{ij}) + \cot(\beta_{ij}))$$

where  $\|e_{ij}\|$  is the length of a primary edge,  $\|e_{ij}^*\|_{\pm}$  is the (signed) length of the corresponding dual edge in the triangle mesh and  $\alpha_{ij}$  and  $\beta_{ij}$  are the two angles opposite to the edge  $e_{ij}$ . (1 Point)

2. Please give an example of a triangle mesh where: (1 Point)
  - (a) at least one weight of the cotan-Laplacian operators is negative;
  - (b) the computation of the cotan-Laplacian operators becomes numerically instable. (1 Bonus Point)
3. Sketch and describe in *detail* what you have to do for the following operations in context of mesh optimization:
  - (a) split an edge; (1 Bonus Point)
  - (b) collapse an edge. (1 Bonus Point)
4. Describe at least two methods to approximate vertex normals of triangle meshes and discuss their advantages and disadvantages. (1 Bonus Point)

<sup>1</sup>C. Loop, *Smooth subdivision surfaces based on triangles.*, M.Sc. thesis, Department of Mathematics, University of Utah, 1987.

<sup>2</sup>Kobbelt, L. (2000).  $\sqrt{3}$ -subdivision. In Proceedings of the 27th annual conference on Computer graphics and interactive techniques - SIGGRAPH '00 (pp. 103–112). New York, New York, USA: ACM Press.